SET-B

GLA UNIVERSITY,MATHURA

COURSE:B.TECH YEAR:II SEM:III Sec:A,B,C,D Branch:CSE

II Mid Term Sub: Object Oriented Programming(ITE 211)

Section A

Objective Type Questions (Attempt all questions) (8x0.5=4 Marks)

1. **Which keyword is used in java for inherit the features of one class into another.**
   - extend
   - extends
   - implements
   - implement

2. Which type of Inheritance **is not supported by java**
   - Multi level
   - Multiple
   - Single -level
   - Hierarchical

3. **A _____ is similar to a program that has a single flow of control.**
   - Thread
   - Process
   - Interface
   - Object

4. By default , variables of interface are
   - finally
   - private final
   - static final
   - protected final

5. **Dividing an integer by zero is**
   - SecurityException
   - IOException
   - NullPointerException
   - ArithmeticException

6. **The catch () is passed _____parameter.**
   - 3
   - 1
   - 0
   - N

7. **Which of the following class can be sub-classed only but can't be instantiated?**
   - Abstract class
   - Final class
   - Virtual class
   - protected class

8. **What we use for comparing two strings for equality**
   - ==
   - equal()
   - =
   - equals()

Section B

Short Answer Type Questions (Attempt any two) (3x2=6 Marks)

Q1. What is a constructor discuss its special properties.

Q2. How do we add a class or interface to a package.

Q3 .What is the two way of creating a thread? What is the difference between Process and Threads?

Section C

Long Answer Type Questions(Attempt any two) **(2x5=10 Marks)**

Q1. Discuss exception handling.WAP in java to generate arithmetic exception and array index out of bound exception and then incorporate necessarily catch blocks to catch and handle them appropriately.

Q2. WAP in java to implementing an interface that calculate the area of a rectangle and circle.

Q3. Explain the method Overloading & Method Overriding with the help of suitable example. Differentiate method Overloading and Method Overriding.

GLA University, Mathura **SET-A**

II-Mid Term Examination, 2012-13

Course: - B.Tech (CS) **II-Year, III-Sem.**

Subject: - Computer Organization (CSE 212) **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:-20**

Notes:-

1) *Answer all questions; Section A is* ***compulsory, any two*** *from Section B and* ***Any two*** *from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

SECTION-A

Objective Questions (Write the correct answer) : **(8*0.5 = 4)**

I. The stack is a specialized temporary …… access memory during ….. and …… operations
- random, store, load
- random, push, load
- sequential, store, pop
- sequential, push, pop

II. A computer system consists of 64 registers of 32 bits each. How many multiplexers are required in bus system?
- 64
- 32
- 2^{32}
- 2^{64}

III. Let SP=0000000 in the stack. How many items are there in the stack if Full=1 and Emty=0?
- 64
- 128
- 256
- 32

IV. A computer system consists of 64 registers of 64 bits each. What is the size of multiplexer in bus system?
- 64*1
- 2^{64}*1
- 6*1
- 4*1

V. Consider an n+k bits one address instructions with k-bits operation code and n-bits address. This instruction format allow maximum ……operation code and ……..addressable memory words.
- 2^k, 2^{n+k}
- 2^k, 2^n
- 2^{n+k}, 2^{n+k}
- 2^{k+1}, 2^{n+1}

VI. Assume a computer with 4-address instruction format, the number of bytes required to encode an instruction is (Assume each address require 24 bits and 1 byte is required for operation code)
- 12
- 73
- 13
- 72

VII. Pipeline improves CPU performance due to
- Reduced memory access time
- parallelism
- Increased clock speed
- None of these

VIII. Consider a case where 4 –segment pipeline with clock cycle time 20 ns in each sub operation to execute 100 tasks. Pipeline system will take how much time to complete the tasks?
- 2060ns
- 20ns
- 2000ns
- 1901ns

SECTION-B (Attempt any two) **(3*2=6)**

I. What do you mean by micro-operation? Discuss various types of micro-operation in brief.

II. A computer uses a memory unit with 128K words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four fields: an operation code field, a mode field to specify one of the five addressing modes, a register address field to specify one of the 100 processor registers, and a memory address. Specify the instruction format and the number of bits in each field if the instruction is in one memory word. How many maximum operations system can perform?

III. An instruction is stored at location 151 with its address field at location 152. The address field has the value 300. A processor register R1 contains the number 200 and Base register contains the number 100.Evalulate the effective address if the addressing mode of the instruction is

a) Direct
b) Immediate
c) Relative
d) Register indirect
e) Index with R1 as the index register
f) Base register

SECTION-C (Attempt any two) **(5*2=10)**

I. Explain various addressing modes for computer system with an example.

II. Why control unit is necessary in computer organization? Write types of control unit and explain any one of them.

III. Evaluate given expression by using general register organization with two address instruction format and stack organization.

$X = (A + B - C * (D - E * F)) / (G * H + K)$

SET-D

Uni. Roll No:-.....

GLA University, Mathura

Course: -B.Tech. II-Year, II-Mid Term (Odd Sem.) Examination, 2012-13

Subject: - Data Structures and Algorithm (CSE211)

Time:-90 Minutes M.M:-20

Notes:-

1. Answer all questions from Group A (which is) **compulsory, Any Two** from Group B and **Any Two** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A (All questions are compulsory) **08X 0.5=04**

Q1. The time required to search an element in a binary search tree having n element is

» $O(\log_2 n)$ » O(1) »O(n) »$O(n\log_2 n)$

Q.2 Consider a sorted binary insertion tree. What must be done to produce a sorted array of numbers from the sorted binary insertion tree?

» In-order traversal »Post-order traversal

» Pre-order traversal »Top-down traversal

Q.3 The number of binary trees with 3 nodes which when traversed in post-order gives the sequence A,B,C is

»3 »7 »9 »5

Q.4 If there exits at least one path between every pair of vertices in a graph, the graph is called

»Complete graph »Disconnected graph »Connected graph »Euler graph

Q.5 A Complete full binary tree with 10 leaves

»Cannot have more than 19 nodes »has exactly 19 nodes

»has exactly 17 nodes »None

Q.6 A 3-ary tree in which every internal node has exactly 3 children. The number of leaf nodes in such a tree with 6 internal nodes will be

»10 »11 »12 »13

Q.7 A binary search tree contains the value 1,2,3,4,5,6,7,8.The tree is traversed in pre-order and the values are printed out .Which of the following sequences is a valid output?

»53124768 »53126487 »53241678 »53124786

Q.8 A 2-3 tree is a tree such that

1. All internal nodes have either 2 or 3 children.
2. All paths from root to the leaves have the same length.

The number of internal nodes of a 2-3 tree having 9 leaves could be

»4 »5 »6 »7

Group-B (Answer any Two) 3x2=6

Q.1 Draw a B-tree of order 3 for the following sequence of keys:
2, 4, 9, 8, 7, 6, 3, 1, 5, 10

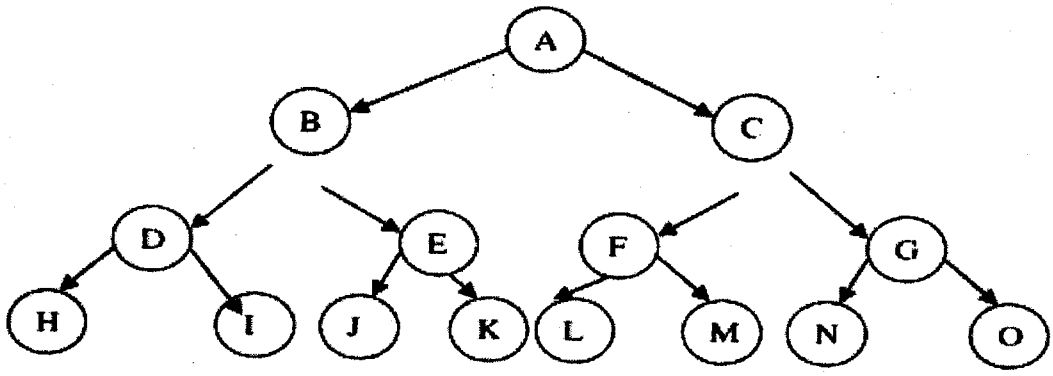
Q.2 Construct a binary tree whose nodes in inorder and preorder are given as follows:
Inorder : 10, 15, 17, 18, 20, 25, 30, 35, 38, 40, 50
Preorder: 20, 15, 10, 18, 17, 30, 25, 40, 35, 38, 50

Q.3 Create an AVL Search Tree by inserting the following elements in the order of their Occurrence 64,1,44,26,13,110,98,85

Group-C (Answer any Two) 5x2=10

Q.1 Write an algorithm to delete a given item from a binary search tree.

Q.2 Show the result of running BFS on the following graph. Show the status of the data-structure used at each stage.



Q.3 Describe the structure of a B+ tree. Draw a diagram showing a B+ tree of degree 2 where the tree:
a. has only two levels (the root node and leaf nodes); and
b. all the nodes are full.
Show the tree which results when one more data item is added to the tree.

ALL THE BEST

GLA University, Mathura
II-Mid Term Examination, 2012-13

SET-A

Course: - B.Tech. (CS) **II-Year, III-Sem.**
Subject: - Discrete Mathematics (CSE-213) **Uni. Roll No:-**
Time:- 90 Minutes **Total Marks:-20**

Notes:-

1) *Answer all questions from Section A, **Any two** from Section B and **Any two** from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

SECTION-A

Objective Questions (Write the correct answer) : **(8*0.5 = 4)**

I. A house has 4 doors and 10 windows. In how many ways can a burglar rob the house by entering through a window and exiting through a door?
- 14
- 40
- 20
- 10000

II. How many arrangements can be made of the letters of the word 'APPLE'
- 32
- 24
- 60
- 120

III. How many 4-digits numbers can be formed by using the digits 2, 4, 6, 8 when repetition of digits is allowed?
- 128
- 16
- 256
- 64

IV. There are 6 gentleman and 4 ladies to dine at a round table. In how many ways they can seat themselves so that no two ladies are together?
- 1800
- 43200
- 10800
- 259200

V. A coin is flipped ten times where each flip comes up either heads or tails. How many possible outcomes contain the same number of heads and tails.
- 252
- 504
- 176
- 45

VI. Find the minimum number of students in a class to be sure that four out of them are born in the same month.
- 12
- 37
- 36
- 48

If S(x) = x is a student and W(x) = x is wise, then what will be the symbolic representation of the following statements:

VII. All students are wise.
- $\forall x: S(x) \rightarrow W(x)$
- $\forall x: S(x)$
- $\exists x: S(x)$
- $\exists x: S(x) \wedge W(x)$

VIII. There are some students who are not wise
- $\exists x: S(x) \rightarrow \neg W(x)$
- $\exists x: S(x) \wedge \neg W(x)$
- $\exists x: S(x)$
- $\forall x: S(x) \wedge \neg W(x)$

SECTION – B (Attempt any two) **(3*2=6)**

I. Using mathematical induction show that $n(n+1)(n+2)$ is divisible by 6 for all $n \in N$.

II. Prove by contradiction method that $\sqrt{5}$ is irrational.

III. Prove the validity of following sentence using rules of inference:
If rents of hotels in Shimla are fixed or price of commodities is reduced, then the income of businessman shall decrease. If the income of businessman decreases, then the farmers shall feel happy. The farmers never feel happy. Therefore the rent of hotels are not fixed.

SECTION – C (Attempt any two) **(5*2=10)**

I. Determine PCNF of $(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$.

II. Prove that $[((p \vee q) \rightarrow r) \wedge (\neg p)] \rightarrow (q \rightarrow r)$ is a tautology without using truth table.

III. Rewrite the following argument using quantifiers, variables and predicate symbols
- All birds can fly.
- Not all birds can fly.
- Some men are genius.
- Some numbers are not rational.
- No well-behaved person is quarrelsome.

(1)

GLA University, Mathura **SET - C**

Second Midterm Examination, 2012-13

Course: - B.Tech. II-Year, III-Sem.

Subject: - Mathematics III (AHM -201) **Uni. Roll No:-**

Time:- 1 Hrs. 30 Minutes **Total Marks:-20**

Notes:-

1) Attempt **ALL** groups.
2) **ALL** questions of group A are compulsory. Attempt **ANY TWO** questions from Group B and **ANY TWO** questions from Group C.
3) All questions of a group should be answered at one place.
4) Answer should be brief and to-the-point.
5) Any missing or wrong data may be assumed suitably giving proper justification.
6) Figures on the right-hand side margin indicate full marks.

GROUP A (8 x1 = 8 Marks)

Q1. Fourier transform of f(x)* g(x) defined by F{ f(x)* g(x)} =

. F{g(x)*f(x)} .. $\frac{1}{2\pi}\int_{-\infty}^{\infty}\{f(x)*g(x)\}e^{i\alpha x}d\alpha$ ... $\frac{1}{2\pi}\int_{-\infty}^{\infty}\{f(x)*g(x)\}e^{-i\alpha x}d\alpha$ .... F{g(x)*g(x)}

Q2. Fourier complex integral of f (x) is given by

. $\frac{1}{2\pi}\int_{0}^{\infty}\int_{0}^{\infty}f(t)e^{-i\lambda t}e^{i\lambda t}dtd\lambda$ .. $\frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{0}^{\infty}f(t)e^{-i\lambda t}e^{i\lambda t}dtd\lambda$

... $\frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}f(t)e^{-i\lambda t}e^{i\lambda x}dtd\lambda$ .... $\frac{2}{\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}f(t)e^{-i\lambda t}e^{i\lambda t}dtd\lambda$

Q3. If $F_s(u)$ is the fourier sine transform of f(x) then fourier sine transform of f(x) cos ax is

. $\frac{1}{2}[F_s(u+a)+F_s(u-a)]$ .. $\frac{1}{2}[F_s(u+a)-F_s(u-a)]$ ... $[F_s(u+a)+F_s(u-a)]$ .... $[F_s(u+a)-F_s(u-a)]$

Q4. Inverse Z-transform of 1 is

. u(t) .. $\delta(t)$ ... $\mu(t)$ .... $\lambda(t)$

Q5. Fourier cosine transform of e^{-ax} is

. $\frac{a}{a^2-p^2}$ .. $\frac{p}{a^2+p^2}$ ... $\frac{a}{a^2+p^2}$ .... $\frac{p}{a^2-p^2}$

Q6. Fourier sine transform of f '(x), that is, $F_s\{f'(x)\}$ is given by

. $p F_s\{f(x)\}-f(0)$ .. $-p F_s\{f'(x)\}$ ... $-p F_C\{f(x)\}$ .... $-p F_C\{f'(x)\}$

Q7. Z-transform of $f(k)=\begin{cases}\frac{a^k}{\lfloor k} & ; k \geq 0 \\ 0 & ; \text{otherwise}\end{cases}$ is

. $e^{-a/z}$ .. $e^{z/a}$ ... $e^{a/z}$ .... $e^{-z/a}$

Q8. Fourier transform of $f^n(x)$, that is, $F\{f^n(x)\}$ is given by

. $p^n F\{f(x)\}$ .. $(ip)^n F\{f(x)\}$ ... $(-ip)^n F\{f(x)\}$ .... $(-ip)F^n\{f(x)\}$

GROUP B (ANY TWO) **(2X2 = 4 Marks)**

Q 1. Fourier transform of e^{-x^2}, that is, $F\left(e^{-x^2}\right)=\sqrt{\pi}e^{-(p^2/4)}$ then find

(I) $F(e^{-ax^2})$ (II) $F(e^{-4(x-3)^2})$

Q 2. Find the fourier sine transform of $\frac{e^{-ax}}{x}$.

Q 3. Find the Z-tranform of $\cos(\alpha k)$; k ≥ 0.

GROUP C (ANY TWO) **(2X4 = 8 Marks)**

Q1 . Using fourier integral representation show that $\int_0^\infty \frac{\cos ux + u\sin ux}{1+u^2}du = \begin{cases}0 & ; x<0 \\ \pi/2 & ; x=0 \\ \pi e^{-x} & ; x>0\end{cases}$

Q 2. Solve the equation $\frac{\partial u}{\partial t}=k\frac{\partial^2 u}{\partial x^2}$; 0 ≤ x < ∞ subject to conditions

(I) u=0 when t = 0, x ≥ 0 (II) $\frac{\partial u}{\partial x}=-\mu$ (a constant) when x=0 and t > 0.

Q3. Find the inverse Z-transform for K ≥ 0 of (I) $\frac{z}{z^2+7z+10}$ (II) $\frac{8z^2}{(2z-1)(4z-1)}$

GLA University

2nd Mid Term Exam

B. Tech: II Year, 3rd semester [C.S, E.C.]

Ethics & Values (AHE: 201)

Marks: 20 Time: 1.30 Hours

This paper consists of two passages. Carefully read the passages and answer the following multiple choice questions. All questions carry equal marks, i.e. 1 mark each.

Passage: 1

What is the most demanding wish other than freedom? Indeed, it is respect for individuals' freedom. Theories of individual rights make the point clear that each individual has a separate life and different preferences. Individuals' preferences should not be sacrificed for either larger social purposes or the common good. We people not only desire but also value the freedom. By the very human nature we like freedom and do not want to live in the restrictions of laws and rules. Nevertheless, we desire those minimum laws and rules which actually protect ourselves from others. Libertarians define freedom in such a way that freedom for individuals is all or nothing. In other words, they argue individuals must be given complete freedom. However, Libertarians differentiate between individual and collective (social) freedom. According to libertarians, there should not be any kind of limitations on individual freedom. However, there is a condition for collective freedom that one is free to the extent that one must not harm the other. Most interesting feature of the Libertarians' idea of freedom is that one has got complete freedom in a very precise sense that one can, if one wishes so, harm himself or herself. Libertarians advocate for the minimal state theory.

What this theory simply means is that state should not limit the liberty of individuals or a government should not regulate the individuals' choices and actions by propounding certain restrictions and laws. Individuals have got fundamental right to liberty. Government has got only role to play in individuals' lives, and that is to protect them from any kind of external threat like theft, fraud, and corruption. Libertarians object the excessive use of the powers of government in the way it intervene in individuals' lives and coerce them to follow certain laws. There are three such objections by Libertarians against the governments. First, government should not make paternalistic laws. Second, government should not make laws for the morals legislation of individuals' lives. Third, government should not redistribute wealth by putting taxes on individuals' salary. By paternalistic laws we mean such laws which are actually for the individuals' safety and protection like the helmet, seatbelt driving safety laws, and the pension laws. Libertarians argue that enforcing such laws on an individual a government undermines his or her risk taking ability. Libertarians believe that government should not propose a certain kind of values or principles for individuals' lives. An individual must be given full freedom to choose his or her own patterns and ways to live the life. However, Libertarians put one condition on this that one's moral choices must not affect the others. On the issue of taxation, Libertarians contend the government and say: by putting taxes on individuals' income government takes away the ownership of an individual. The amount we pay as taxes becomes slavery for an individual. What we earn through our labor talent or skill is our self expression. In this way, a self becomes slave and one's labor becomes forced labor. This is

the reason governments should not redistribute the wealth by following the mechanism of taxation. However, governments can have a check on individuals' property rights. A government must insure fair dealings amongst the individuals in terms of the initial holding of property, justice in acquisition and transfer of money, and other property rights. Libertarians by limiting the excessive role of government maintain: A government should respect an individual's personal life. The life of an individual must be respected as an end and never as means. Respect an individual's life projects independent of collective common goods. And, by respecting an individual a libertarian simply means: government should not interfere in individuals' personal life for accomplishing the purposes of society at large.

Questions:

1) Libertarian objects Paternalistic laws because:
   - Such laws undermine risk taking ability of an individual.
   - Individuals do not have freedom to break such laws.
   - Such laws are not freely practiced.
   - Such laws try to protect individuals and maintain that individuals should not misuse freedom and do not harm themselves.
2) Theories of individual rights point out the fact that:
   - Preferences of an individual should not be sacrificed for larger social purposes.
   - Each individual has a separate life.
   - Each individual has his/her own different preferences.
   - An individual can snatch the rights if they are not provided.
3) Libertarians argue:
   - Individual freedom is all or nothing.
   - Liberty is the fundamental right of an individual.
   - An individual cannot harm oneself.
   - Individual freedom is limited only in the sense that one is not free to harm other.
4) Libertarianism is a naturalistic account of freedom because:
   - It defines the idea of individual freedom on the basis of human nature.
   - It rejects other's freedom.
   - It maintains that an individual desires freedom and does not like rules and restrictions.
   - It argues individual freedom is grounded in rationality.
5) Morals legislation means:
   - Prescribing certain values for individuals to practice.
   - Limiting certain choices of individuals which disturb socio-cultural values.
   - Providing complete freedom to individuals.
   - Allowing some indecent ways of individuals' life.
6) Government has no right to intervene in one's personal choices and actions unless one does not harm the other. In other words, government should not limit the liberty of an individual. This is the idea what libertarian calls
   - Respect for Individuals
   - Respect for freedom of an individual
   - Minimal State theory
   - Maximum individual right theory

7) What we earn through our labor, talent and skill is our property and government has no right to put taxes on it. Libertarians object taxation but they argue that government must insure
   - Just acquisition of money
   - Just transfer of money
   - Just initial holding of property
   - Equal taxes on money and property

8) For a libertarian respect for an individual means:
   - An individual must be treated as an end.
   - An individual must not be treated as a means to some other larger social purposes.
   - Individuals' life projects should not be sacrificed for common good.
   - Sometimes we can treat an individual as an end.

9) Government should not put taxes on individuals' salary because:
   - It takes ownership of an individual.
   - Money is precious for an individual's life.
   - One's profession becomes slavery.
   - The amount we pay as tax is a kind of forced labor.

10) Libertarians would allow Kidney selling and Euthanasia (assisted suicide) on the grounds that;
   - An individual has complete freedom to oneself.
   - We should respect one's personal decision if it does not harm any other.
   - Everyone owns himself or herself.
   - We own others.

Passage: 2

Immanuel Kant, the father of human rights, explains the ideas of individual freedom and respect for human beings in so neat and scientific manner. Being contemporary of Sir Isaac Newton, Kant was very much influenced with Newtonian physics, and he was in full agreement with Sir Newton on the point that nature is governed by the deterministic physical laws. However, this determinism was a sort of challenge for Kant to define, that if we are governed by the deterministic laws then, in what sense we can be said to be free. Kant by respecting the Newtonian Physics initiates his argument in such a beautiful manner and says that there are two kinds of world: 1) Physical, and 2) Moral. Further, he argues that we people are governed by the physical laws in so far as we operate in nature (we operate under law of gravitation, in a certain atmosphere consisting of density, pressure and other physical conditions). However, Kant argues this could be a limitation on the animal life because there exists the one and only physical world for them. But, for individual persons, there exists what Kant refers to as moral world. Moral world is such a world that we create for ourselves. And, we can create it by the virtue of human rationality. Moral world is a world of rationality in which the human life is governed by reasons. Kant says it is only human beings who have got mind and its faculties of thinking and reasoning. By having recourse to these faculties of mind, an individual person can intend, resist and can do certain acts and can refrain herself/himself from doing something. Kant argues that a person is free, and

free by reason. For example, let's compare two persons one is guided by reason and another who does not use rationality. You can easily understand that the one who follows the reason can explain many events of life but who does not follow would be depending either on the other or would be always operating under certain fear or external command. This is the reason why Kant argues that it is rationality through which we get the freedom. Thus, Kantian idea of individual freedom, unlike the Libertarians, is grounded in human rationality. But Kantianism and Libertarians unanimously argue that individual choice should not be sacrificed for larger social purposes. Establishing the idea of individual freedom Kant goes ahead to explain the idea of respect for the human beings. Despite, human beings possess rationality and autonomy they must be respected because they have the capacity to legislate morality for themselves. They can create moral rules for their own lives. This is what Kant calls self-legislation. In other worlds, we can say one can realize the sense of freedom in creating good rules for one's betterment of life.

Kant further tries to explain that morality should be universal and uniform, and should not depend either on personal biasness or external pressure. Morality must be objectively rational in its character that we people can easily understand it without involving the personal reasons, and if we really want to include our personal reasons then we must not personalize the reasons but rationalize them to the extent that they become unbiased, universal and uniform. Kant after explaining the ideas of rationality and freedom further moves to explain the idea of moral accountability and says that we people can be held responsible once we act freely and rationally. However, he makes a very interesting point that it is very much possible that a person does something 'X' and 'X' is right and s/he knows very well that 'X' is right but there is possibility that 'X' is not moral. Really interesting moral fact it is, Kant says that a person who does 'X' but not by his or her moral choice. The person may not have chosen that action or s/he was not free to do any other action. It means the person has performed the action either on the other's choice or under coercion, pressure or force. The action may be right but in Kantian definition it is not moral. So, you do something right but if not on your own choice then your action does not have any moral worth. It cannot be said morally a praiseworthy action. Thus, we can praise or blame someone for his or her actions only in that someone has performed an action with freedom and reason. Kant says moral responsibility can be ascribed to someone who is both a free and rational creature. It is in this sense Kant emphasizes that morality should be free from religion. Kant says morality is an expression of oneself. Morality must freely evolve in oneself. Kantian description of morality is based on the deontology. Deontological morality is a framework in which moral worth of an action, duty, rule, decision or person lies in the very nature of the action, rule, duty, decision or person.

Unlike, Consequentialism, Kantianism focuses on the means, ways, or the procedures we follow to perform an action. This view holds we should only care the actions not the consequences. We should be careful to the means that we choose to fulfill an end and there is no need to look at the ends. Kant says that even if your right means do not produce the right end do not get depressed as your action carry a moral worth within itself. For example, suppose if someone being student honestly study hard by practicing his duty somehow does not get good marks, then this student, according to Kant, need not be worried and disappointed in so far as his action is morally correct and good. Kant put it in this way "Duty for the sake of duty." Kant maintains

universality in his morality by explaining the idea of categorical imperative. He argues in all moral rules there is an imperative but it can be of two kinds: 1) Hypothetical, and 2) Categorical. Hypothetical imperative means the imperative or moral command of an action depends on external conditions. For example, if you want to get good marks then study hard. In this example for a student studding hard is a morally correct thing but it has to qualify a condition, and that is to get the good marks. Categorical imperative is supreme principle of morality. However, categorical imperative means the imperative or moral command of an action is unconditional. For example, do your duty. Kant argues the moral worth of a moral action does not depend on the consequences it produces but depends on our intentions behind our actions. For example, a murderer and a doctor both uses a knife but with different intentions one uses to kill a life and other uses to save a life. Here Kant introduces his beautiful idea of good- will and says that no action can be said to have moral worth unless performed by a sense of good will. What is good-will? Kant replies this question so neatly and says that a will is different from desire. Desires are psychological and represent our emotions and passions. And, they lack determination that is why our desires get changing all the time. However, will is not purely psychological, it is based on reasons and have got determination as well. Kant says, although will is rational and has got strong determination but without being good a normal will may carry wrong intentions. For example; a strong willed person wants to use his determination in killing someone. This is the reason why Kant argues that our will must follow good intentions that's what he calls good-will. He says without good-will morality cannot be performed and all values say knowledge, power and beauty can be misused. For example, a knowledgeable person can misuse the knowledge for the wrong purposes making cheat and frauds, powerful person can misuse the power by beating a week. Without having good-will we cannot act morally.

Questions:

1) A person 'P' performs an action 'X'. And, 'X' is right. And, P knows that 'X' is right. But, his action 'X' cannot be said to be moral because:
   - 'X' is not a moral action.
   - P was not free to perform 'X'.
   - P performed 'X' for others.
   - 'X' was not chosen by person P.

2) Kant argues that the moral world is:
   - Governed by reasons.
   - Determined by physical laws.
   - A world of moral reasons.
   - Governed by rational persons.

3) Kant says human beings are free because:
   - They are rational creatures.
   - They are free by nature.
   - Being rational creatures humans can practice reasons.
   - They have power to exercise their freedom.

4) Kant argues we should respect all human beings:
   - All human beings like to be respected.
   - They are rational beings and end in themselves.
   - They are not means like objects and other thing
   - They can rationally self legislate morality.

5) Someone can be held morally responsible for an action only if :
   - S/he is free to perform an action.
   - S/he is rational creature.
   - S/he has chosen the action by his or her own will.
   - S/he did a right thing.

6) According to Kant morality is:
   - An expression of oneself.
   - It should evolve in oneself through reasons.
   - It should be free from religion.

- It should be free from any kind of coercion, force or pressure.

7) Good-will is the only thing that can be said to have moral worth in genuine sense because:
   - Good-will carries good desires.
   - Good-will carries good intentions.
   - Good-will is governed by good reasons.
   - Good-will is different from other wills.
8) Categorical imperative is the supreme principle of morality. Which of the following statements are not correct about categorical imperative:
   - The moral command in categorical imperative is unconditional.
   - The moral command in categorical imperative has nothing to do with the consequence, result or end of an action.
   - The moral command in categorical imperative will bring maximum good.
   - The moral command is chosen by God.
9) which of the following statement is/ are correct for deontological morality framework:
   - Moral worth of an action lies in the nature of an action.
   - Moral worth of an action lies in the results it produces.
   - Moral worth of an action lies in the actor.
   - Moral worth of an action lies in the motivation for an action.
10) Kantianism and Libertarianism both are altogether different theories but they have some common agenda.
   - Both advocate for rights of individuals.
   - Both explain the ideas of individual freedom and respect for human beings.
   - Both argue that individual choice should not be sacrificed for larger social purposes.
   - Both define individual freedom on the basis of human nature.